CSA1712

ARTIFICIAL INTELLIGENCE

NAME: VARSHINI M

REG NO.: 192511215

1. Answer: C) Greedy Best-First Search

Explanation: It selects the node with the lowest heuristic value **h(n)** without considering the path cost.

2. Answer: C) f(n) = g(n) + h(n)

Explanation: A* combines the actual path cost **g(n)** and heuristic estimate **h(n)**.

3. Answer: B) Admissibility

Explanation: An admissible heuristic never overestimates the actual cost, ensuring optimal solutions.

4. Answer: B) Hill Climbing

Explanation: Hill Climbing improves the current solution step by step using local information.

5. Answer: B) Minimax Algorithm

Explanation: Minimax chooses the best move by considering both the player's and opponent's actions.

6. Answer: B) Greedy Best-First Search

Explanation: It uses only the estimated distance to the goal, ignoring the path cost.

7. Answer: B) A* Search

Explanation: A* finds the shortest path using both travelled cost and estimated remaining cost.

8. Answer: B) Constraint Satisfaction Problem

Explanation: Exam scheduling requires satisfying all constraints such as no clashes and limited rooms.

9. Answer: C) Minimax Algorithm

Explanation: Minimax evaluates all possible moves and selects the best one based on utility values.

10. Answer: B) Online Search Agent

Explanation: It gathers information while exploring an unknown environment.

11. Answer: B) It ensures optimality

Explanation: An admissible heuristic guarantees that A* finds the optimal solution.

12. Answer: B) Reduces number of nodes evaluated

Explanation: Alpha-Beta pruning skips unnecessary branches, improving search efficiency.

13. Answer: C) Getting stuck in local maxima

Explanation: Hill Climbing may stop at a local optimum instead of the global optimum.

14. Answer: C) Constraint satisfaction

Explanation: A valid CSP solution satisfies every given constraint.

15. Answer: B) Estimate utility of a state

Explanation: The evaluation function estimates how good a game state is.

16. Answer: A) 12

Explanation: $f(n) = g(n) + h(n) = 7 + 5 = 12$.

17. Answer: B) A* becomes non-optimal

Explanation: If the heuristic overestimates, A* may not find the shortest path.

18. Answer: C) 31

Explanation: Total nodes = $2^0 + 2^1 + 2^2 + 2^3 + 2^4 = 31$.

19. Answer: B) 27

Explanation: Leaf nodes = $3^3 = 27$.

20. Answer: B) O(b^(d/2))

Explanation: In the best case, Alpha-Beta pruning reduces Minimax complexity to **O(b^(d/2))**.